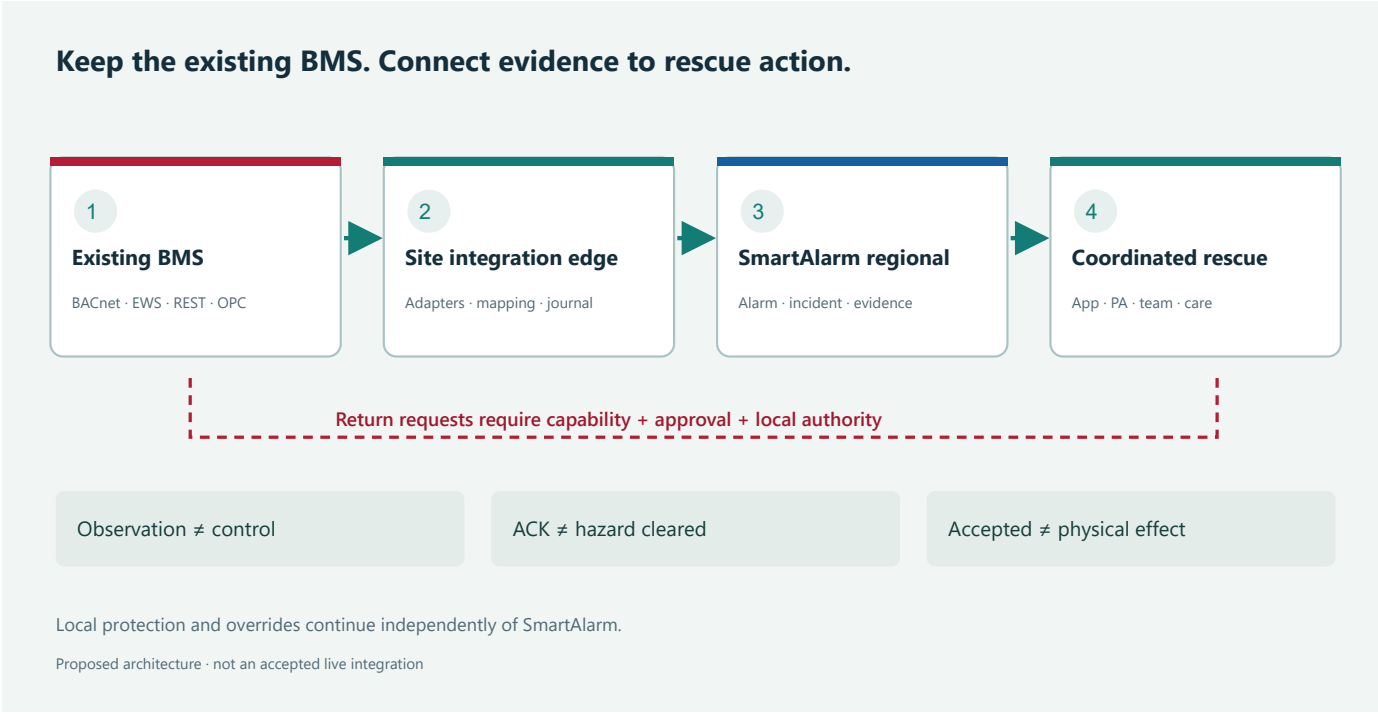
Connect existing BMS to coordinated rescue

Research and extension design · 24 October2026 · International, multi-tenant and multi-industry scope.

Baseline 3.0 integrates a lifecycle-managed BMS integration domain with a shared connector platform and separate profiles for each product, version, interface and license. The BMS retains building-services operation and local control authority; SmartAlarm uses its evidence for hazard assessment, evacuation guidance, public address, proactive assistance and rescue coordination. Research covers **8 vendor/product families and 11 proposed connector profiles**, without treating protocol support as whole-brand compatibility. This package specifies **40 FR-BMS requirements,8 new screen contracts,20 business acceptance cases and 28 protocol cases**, together with a schema and JSON profiles. Their structure has been validated, but **adapters are not implemented and no live BMS has been connected**; activation requires qualification of the exact deployment. The integrated baseline contains 241 screen mockups, 37 workflows and a 188-slide proposal in each language. The eight BMS screens are integrated into the mockup catalog. The API contracts below are canonical internal SmartAlarm contracts, not vendor endpoints. This remains an implementation design package, not delivered BMS driver software.



1. Compatibility belongs to a product and operation

No customer product or version is known, so the architecture must accommodate modern API-enabled BMS, supervisors exposing BACnet/OPC, and legacy systems through an approved gateway or register map. Their information fidelity differs: point reads do not establish an alarm lifecycle; an alarm list does not establish acknowledgement; writable points do not authorize safety-related effects. Operational capability is the intersection of **implemented ∩ vendor-supported ∩ licensed ∩ site-authorized ∩ accepted**. Protocol fallback is never automatic because the alternate path can change identities, authority and semantics.

Profile	Product and interface	Evidence and qualification
`bms.siemens.desigo-wsi`	Desigo CC REST / WSI subscriptions	V8 specification describes REST read/write, events and history. WSI authentication has version-specific session/heartbeat behavior. Verified northbound OPC is DA, not UA. Qualify project/extension, HTTPS and PICS for a BACnet binding. Siemens WSI
`bms.schneider.ebo-ews`	EcoStruxure Building Operation SOAP EWS / SmartConnector REST	The SmartConnector2.6/EWS1.2 guide was tested against EBO7.1.2. EBO3+ uses SoapEwsRestProvider. Do not assume root discovery or bulk support. Acknowledgement targets an alarm event, not every alarm-state transition. EBO EWS
`bms.jci.metasys-rest`	Metasys Version-pinned REST / SSE	API v6 documents objects, command schemas, alarms, histories and SSE. Inspect each batch result rather than HTTP200 alone. Monitoring and Commanding needs separate licensing; the Metasys name alone does not establish compatibility. Metasys API
`bms.honeywell.ebi-api`	Enterprise Buildings Integrator Vendor SDK / web services	EBI2025 advertises enterprise web services, but the public page does not provide an implementation contract. Older R600 guide specifications discuss OPC DA/A&E/HDA and APIs; they do not establish equivalent REST or UA capability in EBI2025. EBI
`bms.honeywell.webs-niagara`	WEBs N4 Niagara OEM station interface	WEBs is Niagara-based and needs an OEM profile. Modules, station exports, permissions and licensing determine capabilities. Do not reuse EBI APIs. WEBs N4
`bms.tridium.niagara-local`	Niagara local station / Supervisor Qualified oBIX / licensed module	Supervisor4.6 documentation and the 2025 driver catalog identify oBIX. A marketplace REST module is a separate capability, not universal built-in REST. Alarm acknowledgement requires a qualified contract. Niagara local
`bms.tridium.nds-cloud`	Niagara Data Service Cloud API	NDS is a separate cloud service for histories/tags, live read/write and alarm queries. Prerequisites include SMA and a reseller. Alarm acknowledgement is unverified; WAN cloud is not a critical local-control fallback. Niagara Data Service
`bms.delta.enteliweb`	enteliWEB Vendor API / BACnet Web Services	The 4.23 catalog, updated November2022, lists an -API add-on. Confirm current entitlement; a brochure is insufficient for authentication, endpoints and lifecycle implementation. enteliWEB
`bms.alc.webctrl-events`	WebCTRL event/telemetry integration Outbound GET/POST / MQTT add-on	The 2025 sheet mentions alarm REST and outbound alarm-triggered GET/POST; MQTT is an add-on. Outbound events do not prove REST point commands or native acknowledgement. Initial scope is event/telemetry intake. WebCTRL

Profile	Product and interface	Evidence and qualification
`bms.alc.webctrl-soap-sdk`	WebCTRL SOAP / SDK SOAP / licensed Java SDK	SOAP needs a version-specific method contract; JavaSDK licensing is case-by-case. Qualify read, acknowledgement, commands and history by method and account scope. WebCTRL SDK
`bms.distech.ecnet-niagara`	EC-Net Facilities / EC-Net4 Niagara OEM station interface	EC-Net Facilities is the renamed Niagara-based EC-Net4. ECLYPSE REST belongs to the controller family, not automatically the EC-Net Supervisor. Qualify OEM modules and station access. EC-Net

Vendor evidence and applicability

Table links open vendor documentation directly. Named versions identify researched documents, not a SmartAlarm-tested support range.

Read each vendor entry with its product boundary. In particular, Metasys licensing is a separate commercial prerequisite, the older EBI R600 guide specification does not prove every feature is implemented in EBI2025, and ECLYPSE controller REST is distinct from an EC-Net supervisor. Historical application-note limits are not current capacity commitments. Do not quote unsupported prices, rate limits, token lifetimes, point limits or development durations. Metasys licensing · EBI R600 · Distech developer portal

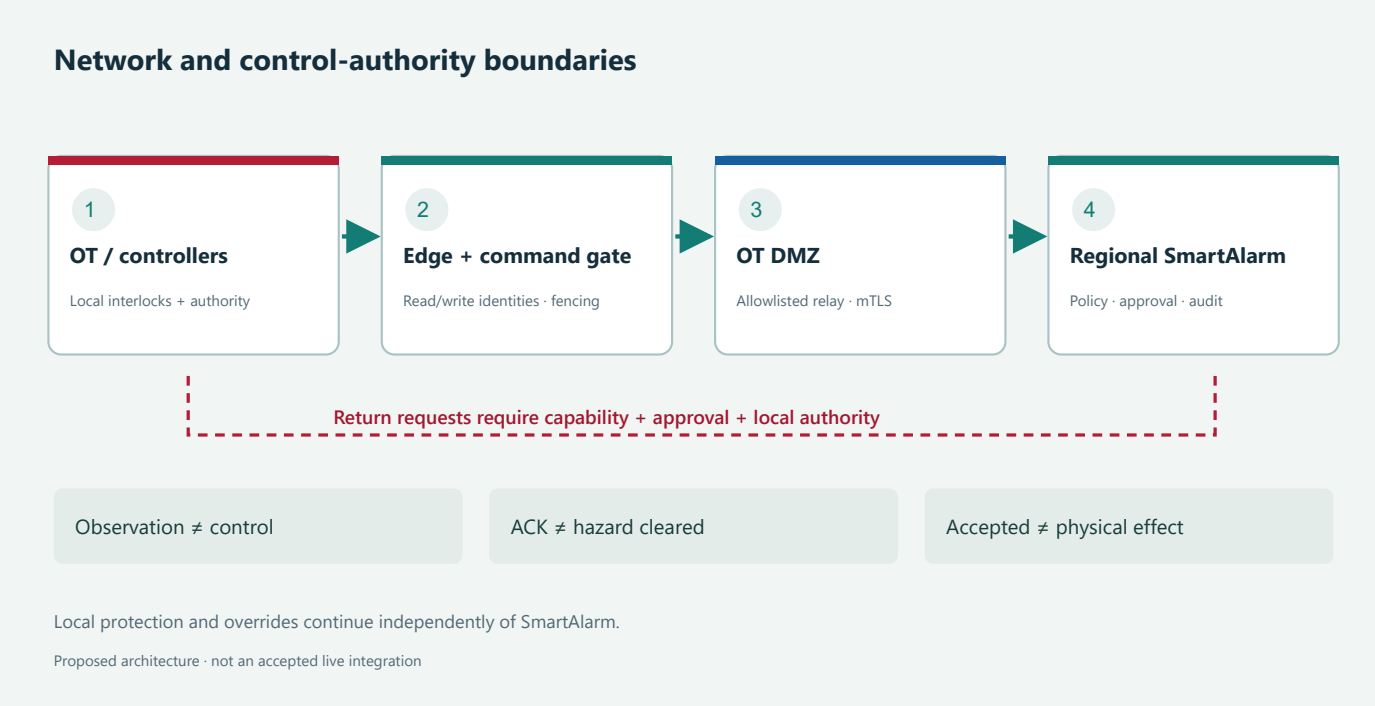
Protocols provide transport and models, not business authority

Layer	Design implication	Source
BACnet/IP / MS/TP	Read the exact-model PICS. COV is value change, not the alarm lifecycle. An IP router does not give an MS/TP bus IP-level capacity. Bound discovery and polling.	PICS
BACnet/SC	WebSockets/TLS protect transport; manage CA, trust, expiry and hub failover. SC does not grant control authority or replace OT segmentation.	BACnet/SC
BACnet priority	There are16 priority levels; NULL relinquishes one slot to lower priority/default. Application TTL and ownership are not guaranteed. Edge power loss can leave a slot retained.	Priority model · Relinquish
OPC UA	Preserve StatusCode, source time and server time. A&C and history are separate capabilities. ConditionRefresh loads retained conditions; it is not history replay and can race with new events.	DataValue · A&C refresh
OPC Classic	DA, A&E and HDA are not OPC UA. An approved Windows bridge belongs inside OT; do not expose DCOM to the Internet.	Desigo OPC

Layer	Design implication	Source
Modbus	Require the exact model/firmware register map, PDU address, type, signedness, word order, scaling, units and invalid sentinels. Register-derived alarms must be marked derived. Modbus Security is separate.	Modbus · Security
MQTT / webhook	Payload and alarm identity are project contracts. QoS does not prove physical effect; deduplication remains necessary. Do not retain immediate commands. Authenticate webhooks, prevent replay and reconcile state.	MQTT5

Brick/Haystack can support semantic import without forcing replacement of the customer BMS. Separate location, composition, service-flow and point relationships; pin model versions. Keep semantic point identity separate from native references for controlled binding migration. Brick relationships · External references · Haystack points

2. The site edge preserves semantics and control boundaries



Reuse the existing Integration runtime, Physical integration, event bus, Incident, Routing, PA, Care and Command aggregate. The BMS domain adds system dossiers, native-alarm projections, semantic mapping and qualification; it does not create a second Command service. Locate edge components in OT or DMZ according to the network assessment, using approved conduits and preferably outbound mTLS sessions to regional ingress. A relay is not a transparent IP bridge to the field bus. Observation, alarm exchange, commands and operations are four independent contracts; approved monitoring remains useful even without control authorization. NIST OT architecture

Component	Responsibility
Registry / qualification	Product/build/license/interface/evidence; signed profiles; connection success never enables writes.
Read acquisition	Separate subscription, polling and history workers; source budgets, reconnect jitter, cursors and raw evidence.
Semantic mapping	Immutable revisions for type/unit/enum/quality and equipment/zone/route dependencies.
Edge journal	Durable commit, inbox/outbox, gaps, checksum and epoch; reserve alarm capacity without silent drops.
Alarm bridge	Native occurrences, snapshot+stream, external ACK intents, provenance and anti-echo.
Command gate	Separate identity, allowlists, approval digest, expiry, mapping revision, local permissives and fencing.
Readback / health	Reconcile outcomes; monitor subscription, freshness, clock, queue, license/certificate and affected zones.

Each decision retains its proper authority

Authority	Data and boundary
BMS/controller	Native objects, raw values, native alarms, local override and winning priority.
Site-approved mapping	Equipment/space/zone semantics, physical lineage and relationships approved for routing.
SmartAlarm	Incident, SOS, tasks, assistance profiles, approval/audit and Command intents.
Qualified feedback source	Physical outcome evidence; command=true does not prove fan flow, unlock does not prove door open.

Integration does not grant the BMS access to raw medical records, ages or personal identities. Writeback contains only the permitted incident reference, zone, severity and minimal response state. Occupancy counts, badge passage and location traces are quality-qualified presence evidence, not proof that a named person remains inside or is safe. An LLM reads scoped projections with sources and timestamps; approved response rules do not depend on LLM output, and the LLM cannot write to the BMS.

HA, offline and relinquishment are enforced at dispatch

Partition by tenant/site/connector and allow one active writer per control domain. Fencing must prevent the old edge from actually writing, not merely elect a cloud leader. Where a legacy controller cannot support sufficient safeguards, read-only HA remains viable while writer failover needs a local procedure. Ledger deduplication and idempotency do not make physical actuation exactly-once. On restart, query/reconcile outstanding commands and priority slots before enabling writes. Never replay expired commands.

A Command TTL prevents late dispatch; it does not guarantee that BACnet will remove an already written value. A powered-off edge cannot promise to send NULL on time. The site must accept a controller-level fallback/ownership arrangement or local operator handover; otherwise that capability remains disabled. Relinquishment releases an authorized slot; it is not writing OFF or restoring an old value. Unproven ownership requires manual reconciliation.

3. Functions cover the entire connection lifecycle

The following requirements are integrated in the existing FRD with inputs, preconditions, processing, durable outputs, failure behavior, permissions and acceptance. Canonical workflows are WF-34 onboarding and qualification; WF-35 alarm reconciliation and rescue; WF-36 governed commands; WF-37 lifecycle and recovery. WF-01–33 are preserved. Every row inherits tenant/site authorization, audit, actual/exercise isolation and version guards.

ID	Function	Acceptance condition	Screens
FR-BMS-01	Site BMS inventory	Activation requires build, interface and evidence.	BMS01, I01–I02, F17
FR-BMS-02	Capability-specific access	Read, history, ACK and commands have separate server permissions.	I02, I07, F03
FR-BMS-03	Bounded discovery	Scan approved scope only; cancellable; alarm intake remains available.	I02, S08, BMS01
FR-BMS-04	Point and object import	Preview row errors, duplicate IDs and encoding; commit a revision.	F04, S06
FR-BMS-05	Unit and semantic mapping	Golden fixtures verify scale, enums, types and byte order.	I03, F04
FR-BMS-06	Equipment and zone topology	Distinguish room, HVAC zone, fire compartment and route edge.	BMS02, F05, S02–S04
FR-BMS-07	Mapping publication and impact review	Show dependencies and approve revisions; rollback does not undo physical effects.	BMS05, F04, F17
FR-BMS-08	Observation quality and freshness	Zero differs from missing; late data cannot overwrite newer observations.	F01, F07, S07
FR-BMS-09	Subscriptions and polling	Renew/reconcile lost subscriptions; polling follows measured budgets.	I05, BMS06
FR-BMS-10	History and investigation	Preserve UTC, timezone, gaps and raw/interpolated distinctions.	F07, O21–O22
FR-BMS-11	Native alarm lifecycle	Active, ACK, confirmed and suppressed are independent; unsupported=null.	BMS03, O03, F07

ID	Function	Acceptance condition	Screens
FR-BMS-12	Priority and severity mapping	Do not invert priority numbers globally; test per source.	S11–S13, O05
FR-BMS-13	Correlation and provenance	A BMS mirror and direct panel feed are not independent evidence.	O05–O07, BMS03
FR-BMS-14	Controlled external ACK	Target the correct occurrence/version; failed receipt remains pending.	BMS03, O03, F07
FR-BMS-15	Observe external acknowledgement	Preserve source actor; ACK does not close the rescue incident.	BMS03, O04
FR-BMS-16	Publish incident summaries to BMS	Approved namespace, minimal data and anti-echo; no medical records.	BMS08, I08–I10
FR-BMS-17	Operator alarm grouping	Retain raw events and grouping counts; unacknowledged P1 remains visible.	O03, BMS03
FR-BMS-18	Maintenance and suppression	Require reason, owner and expiry; elapsed maintenance does not prove restoration.	F14, F18
FR-BMS-19	BMS evidence for evacuation	Stale door/fan/lift evidence lowers confidence or withdraws guidance.	Q04–Q05, C35–C38
FR-BMS-20	Proactive rescue	Combine needs and fresh presence; occupancy counts are not identities.	H08–H11, O11–O12
FR-BMS-21	Approved PA and app response	Do not wait for an LLM; echo/history cannot repeat broadcasts.	O13–O17, P01–P09
FR-BMS-22	Control impact preview	Show target, owner, priority, preconditions and affected zones.	F09–F10, BMS04
FR-BMS-23	Approved commands with expiry	Parameter changes invalidate approval; local override retains priority.	F10, BMS04
FR-BMS-24	Priority lease and relinquishment	Use permitted slots only; never clear an unproven foreign slot.	BMS04, F13
FR-BMS-25	Physical outcome reconciliation	Separate transport receipt, controller report and independent feedback.	F11–F12
FR-BMS-26	Timeout and unknown outcomes	Do not resend unknown effects; query/read back first.	F12, BMS07
FR-BMS-27	Local override and handover	Do not restore old commands on reconnect; assign cleanup ownership.	F13, BMS04, BMS07

ID	Function	Acceptance condition	Screens
FR-BMS-28	Durable journal and offline operation	Do not ACK before durable commit; local protection remains independent.	I05, G02, BMS07
FR-BMS-29	HA and single writer	Fence at the actual dispatch point; standby cannot write.	G03–G04, BMS07
FR-BMS-30	Alarm resynchronization after gaps	Uncertain snapshot/cursor exposes gaps; one page cannot clear missing alarms.	BMS03, BMS07
FR-BMS-31	Schema drift and upgrades	Rename preserves identity; type/unit changes quarantine affected capabilities.	BMS05, G09–G10
FR-BMS-32	License, certificate and account lifecycle	Test rotation; revocation blocks privileges; never log secrets.	I05, I07, F17
FR-BMS-33	Data- and zone-aware health	Online gateway does not hide stale points; alarm latency is measured separately.	I05, F01, BMS01
FR-BMS-34	Capacity, backfill and retention	Per-BMS quotas; historian work cannot starve alarm queues.	BMS06, G02
FR-BMS-35	Actual and exercise isolation	Simulation never sends real ACK/writes; commissioning has a separate plan.	I04, F15–F16
FR-BMS-36	Configuration-specific FAT/SAT	Sign off per capability; passing reads does not authorize commands.	F15–F17, BMS01
FR-BMS-37	Support and forensic export	Timeline includes revisions/sources; redact secrets and health data.	O21–O22, G07–G08
FR-BMS-38	Evidence-based BMS AI assistance	LLM reads scoped, sourced evidence; it never controls directly.	Q06–Q09
FR-BMS-39	Tenant/site isolation and residency	Repeated native IDs remain isolated; cross-site access uses authorized projections.	G01, T02–T04, BMS01
FR-BMS-40	Controlled retirement and handover	Review open alarms/leases, hand over, revoke, verify and retain audit.	BMS01, F13, G06

Building-service subsystems have different actuation boundaries

Subsystem	Design scope
HVAC	Observe modes, temperature, pressure and damper feedback; bounded setpoints only if approved. Comfort control does not replace smoke control.

Subsystem	Design scope
Smoke control	Observe sequence status, fan proof, fault/inhibit. Request only approved controller/cause-and-effect sequences; never assemble a safety sequence from arbitrary outputs.
Fire / water / pumps	Observe alarm, trouble, supervisory, run, flow, pressure and impairments. Monitor by default; reset/silence/start/release needs a separate dossier. Wet-pipe sprinklers are not individually activated through an API.
Doors / access	Observe demand, contacts, faults, overrides and route relevance. Authorized unlock does not establish an open door or safe route.
Elevators	Observe floor/mode/fault/recall as exposed; an available flag does not authorize evacuation by elevator.
Power / UPS / generator	Observe power loss, readiness and affected loads; generator run does not prove every critical load is powered.
Flood / gas	Observe sensors, levels, valve/pump feedback and approved isolation sequences; missing is not zero.

Eight BMS screen contracts integrated into the 241-screen baseline

BMS01 · BMS dossier and capabilities

Item	Contract
Visual	BMS → edge → SmartAlarm
Fields	vendor, product, build, modules, license evidence, interface, owner, capability matrix
Primary action	Review dossier; inspect health; activate accepted capabilities
Specific failure states	license unknown; certificate expired; incompatible build; partial coverage
Permissions	bms.configure / acceptance approver
Existing links	I02, F17, I05

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS02 · Semantic topology

Item	Contract
Visual	controller → equipment → served zone → route edge
Fields	native ID, equipment, room, fire compartment, HVAC zone, relation type, evidence, revision
Primary action	Map; validate relationships; propose revision

Item	Contract
Specific failure states	orphan point; cyclic containment; conflicting zones
Permissions	bms.configure / bms.mapping.approve
Existing links	F04, F05, S15

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS03 · Cross-system alarm reconciliation

Item	Contract
Visual	native condition ↔ incident timeline
Fields	occurrence, generation, active/ack/confirm, source actor, cursor, gap, linked incident
Primary action	ACK exact occurrence; link incident; inspect receipt
Specific failure states	unknown ACK; stale snapshot; recurrence conflict; stream gap
Permissions	bms.read / bms.ack
Existing links	O03, O05, F07, F12

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS04 · Control ownership and priority

Item	Contract
Visual	priority ladder + demand / actual feedback
Fields	target, current winner, priority slots, lease, local override, approved intent, expiry
Primary action	Request authority; relinquish an approved owned slot
Specific failure states	higher-priority override; foreign owner; expired lease; unknown feedback
Permissions	bms.command.request / bms.command.approve
Existing links	F09–F13

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS05 · Point drift and impact

Item	Contract
Visual	before / after diff + dependency graph
Fields	old/new native reference, type, unit, mapping hash, dependent rules/routes/commands
Primary action	Approve or reject mapping migration
Specific failure states	reused ID; changed type/unit; critical rule affected
Permissions	bms.mapping.approve
Existing links	I03, G09, F17

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS06 · Capture budget and freshness

Item	Contract
Visual	zone heatmap + alarm / telemetry / history queues
Fields	poll interval, batch size, subscriptions, queue lag, rate limits, backfill budget, freshness
Primary action	Apply reviewed policy; pause backfill; inspect stale points
Specific failure states	source throttling; queue pressure; partial gaps
Permissions	bms.configure / operations
Existing links	I05, G02, O22

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS07 · Edge recovery and handover

Item	Contract
Visual	writer / journal / cursor recovery swimlane
Fields	owner epoch, source epoch, checkpoint, journal gap, outstanding commands, leases
Primary action	Reconcile; hand over; enable writes after checks
Specific failure states	split brain; missing journal; old epoch; unverified snapshot
Permissions	bms.reconcile / site OT approver

Item	Contract
Existing links	G03, G04, F12, F13

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

BMS08 · Exchange and loop trace

Item	Contract
Visual	origin → BMS → SmartAlarm causation path
Fields	origin event, causation, namespace, native outbound object, hash, receipt, quarantine reason
Primary action	Inspect event path; quarantine looping mapping
Specific failure states	echo; reentry; conflicting identity; unsupported writeback
Permissions	bms.configure / incident reviewer
Existing links	I06, I08, O05

Ready/loading/empty/partial/stale/offline/denied states need appropriate visuals and labels. Unavailable actions show a reason rather than color alone. Separate native raw evidence from translated labels; preserve IDs, units and native codes.

4. Technical contracts preserve meaning across vendors

Four artifacts have independent versions

The adapter image has a digest/SBOM and SDK version; the product profile has exact-version capabilities and evidence; the site binding has points/zones/lineage/mapping revisions; the operation policy has allowlists, approvals, priority, expiry and acceptance. Upgrading one artifact cannot increase the authority of the others. Publishing a catalog profile differs from accepting a site deployment. Included JSON profiles intentionally declare `not_implemented`, unverified site versions, `enabled_capabilities=[]` and `write_enabled=false`; they contain no customer endpoints or secrets.

```
ConnectorFactory.create(profile, secretProvider, clock, durableStore, auditSink)
inspectEndpoint(ReadOnlyProbeContext) -> EndpointIdentity + CapabilityEvidence
importInventory(ApprovedScope, Cursor?) -> InventoryPage
readPoints(PointRefs, ReadPolicy) -> ObservationBatch
subscribe(ApprovedScope, DurableCursor?) -> AsyncStream<NativeEnvelope>
reconcile(ReconcileRequest) -> SnapshotPage + Watermark + Gaps
readHistory(HistoryRequest, Cursor?) -> HistoryPage | Unsupported
acknowledge(AuthorizedAckEnvelope) -> ExternalOperationReceipt
dispatch(AuthorizedCommandEnvelope) -> DispatchReceipt
readback(CommandRef) -> ReconciliationEvidence
relinquish(AuthorizedReleaseEnvelope) -> ReleaseReceipt | Unsupported
health() -> Endpoint + Subscription + Freshness + Queue + TrustHealth
```

```
drain() -> OutstandingIntents + Checkpoint
close() -> CloseResult
```

Unsupported methods return typed errors, never no-op success: `Unsupported`, `PermissionDenied`, `LicenseMissing`, `MappingMismatch`, `StalePrecondition`, `RateLimited`, `TransientTransport`, `SourceRejected`, `OutcomeUnknown`. Adapters preserve native errors/cursors/sessions; orchestration cannot construct raw requests to bypass them. A read-only probe must not test writes against production points to discover permissions.

Entity	Core fields
BmsSystem	tenant/site, vendor/product/build, region/timezone, modules/licenses, owner, interface dossier
BmsConnection	profile, endpoint/secret references, connector group, rate budgets, lifecycle, separate health
BmsObject / PointBindingRevision	native stable ID, canonical physical point, semantic/type/unit/enum/scale, source and approved mapping revision
TopologyRelation	located_in / part_of / feeds_air / feeds_water / serves / controls / exposes / protects; source and approval
BmsAlarmOccurrence	native ID, generation, source epoch, condition/branch/event IDs, active/ack/confirm/retained/suppressed/out-of-service, timestamps
BmsAckIntent	actor, exact occurrence/version, comment, correlation, pending/accepted/rejected/unknown and receipt
ControlLeaseDetail	existing command_id, authorized target/slot/owner, before/desired state, expiry, release policy and physical outcome contract
MappingDrift / SourceAuthorityPolicy	change impact, invalidated capabilities, migration approval; physical lineage and quality-based source authority

Identity, alarms and commands do not share one state flag

Store native alarm dimensions independently: active, acknowledged, confirmed, retained, suppressed and out-of-service; unsupported dimensions are null, not false. Source return-to-normal does not close an incident or establish occupant safety. Internal ACK accepts operator responsibility; external ACK targets the exact occurrence/event generation and needs receipt/readback. A late acknowledgement cannot affect a recurrent alarm. Exact deduplication precedes correlation, using tenant, site, BMS, native alarm ID, generation, source epoch and source event ID where available. Synthetic identity algorithms must declare their limits.

On reconnect, open/buffer the stream, obtain a snapshot and merge against its watermark where supported. Without atomic snapshot/cursor semantics, keep partial/gap status, use source-specific reconciliation and manual review; absence from one page cannot clear an alarm. Historical replay is evidence, not permission to repeat PA, dispatch or expired commands. BMS and direct-panel paths retain their provenance but a shared physical lineage does not count as two independent confirmations.

Commands retain the existing aggregate and states: `requested/approved/dispatched/accepted/executing/succeeded/failed/unknown/expired`. Approval binds target, parameters, mapping and policy revision digests; changes invalidate approval. Transport receipt,

controller execution report and independent physical feedback are separate evidence. Multi-target plans contain child commands, not a presumed all-or-nothing transaction. Compensation is a new authorized command, not automatic physical rollback.

Writeback uses a dedicated namespace and origin/causation/correlation. If the BMS cannot preserve metadata, a bridge ledger links native outbound objects, effect keys and payload hashes; suppress only proven echoes, not every similar text event. Secrets remain vault references. Derive tenant/site authority from authenticated bindings rather than trusting payload tenant_id.

Canonical API and event contracts in baseline 3.0

The existing API contract now includes 23 BMS operations, 67 operations in total. Earlier draft aliases such as `/bms-systems` are superseded by the paths below. Request/response types and guards are normative in API-Contract.json.

Method	Path	Permission	Request → success response
GET	`/bms/connections`	`bms.connection.read`	— → BMSConnectionPage
POST	`/bms/connections`	`bms.connection.create`	BMSConnectionCreate → BMSConnectionView
GET	`/bms/connections/{connection_id}`	`bms.connection.read`	— → BMSConnectionView
POST	`/bms/connections/{connection_id}/discoveries`	`bms.discovery.run`	BMSDiscoveryRequest → BMSJob
POST	`/bms/connections/{connection_id}/map-revisions`	`bms.mapping.edit`	BMSMapDraftCreate → BMSMapRevision
POST	`/bms/connections/{connection_id}/validations`	`bms.mapping.validate`	BMSValidationRequest → BMSJob
POST	`/bms/connections/{connection_id}/qualifications`	`bms.qualification.submit`	BMSQualificationCreate → BMSJob
POST	`/bms/connections/{connection_id}/activations`	`bms.connection.activate`	BMSActivationRequest → BMSJob
GET	`/bms/connections/{connection_id}/points`	`bms.point.read`	— → BMSPointPage
GET	`/bms/connections/{connection_id}/alarms`	`bms.alarm.read`	— → BMSAlarmPage
POST	`/bms/connections/{connection_id}/alarms/{alarm_id}/native-acknowledgements`	`bms.alarm.ack_native`	BMSNativeAckRequest → BMSMutationReceipt
POST	`/bms/connections/{connection_id}/command-previews`	`bms.command.preview`	BMSCommandPreview → BMSCommandPreviewResult
POST	`/bms/connections/{connection_id}/commands`	`bms.command.submit`	BMSCommandSubmit → BMSMutationReceipt

Method	Path	Permission	Request → success response
GET	`/bms/connections/{connection_id}/commands/{command_id}`	`bms.command.read`	— → BMSMutationReceipt
POST	`/bms/connections/{connection_id}/commands/{command_id}/reconciliations`	`bms.command.reconcile`	BMSCommandReconcile → BMSMutationReceipt
POST	`/bms/connections/{connection_id}/reconciliations`	`bms.reconciliation.run`	BMSReconciliationRequest → BMSJob
POST	`/bms/connections/{connection_id}/lifecycle-transitions`	`bms.lifecycle.change`	BMSLifecycleRequest → BMSJob
GET	`/bms/connections/{connection_id}/health`	`bms.health.read`	— → BMSHealth
GET	`/bms/connections/{connection_id}/jobs/{job_id}`	`bms.job.read`	— → BMSJob
GET	`/bms/connections/{connection_id}/policies/request-budget`	`bms.policy.budget.read`	— → BMSRequestBudgetView
PATCH	`/bms/connections/{connection_id}/policies/request-budget`	`bms.policy.budget.change`	BMSRequestBudgetUpdate → BMSRequestBudgetView
GET	`/bms/connections/{connection_id}/policies/incident-export`	`bms.policy.export.read`	— → BMSIncidentExportView
PATCH	`/bms/connections/{connection_id}/policies/incident-export`	`bms.policy.export.change`	BMSIncidentExportUpdate → BMSIncidentExportView

Commands submit an approved preview and approval digest to the existing physical Command executor; no raw point write is exposed. A 202 receipt proves durable local acceptance, not physical success. HTTP 409 covers state/idempotency/authority conflict; 412 covers ETag/evidence revision mismatch; 422 semantic invalidity; 429 budget exhaustion; 503 upstream/durable acceptance unavailable. Unknown command outcomes require reconciliation without blind replay.

Request-budget policy can pause history backfill without pausing active-alarm reconciliation. Incident export requires separately qualified `export_incident` capability and an approved namespace/binding. The only export fields are incident_id, severity, status, zone_id, mode and occurred_at. Echo markers and quarantine-on-echo are mandatory; quarantine disables export. Lifecycle actions use the typed drain/suspend/resume_read/upgrade/rollback/retire contract and expected revision, approval and handback evidence.

Event-Schemas.json

```
com.smartalarm.bms.alarm_changed.v1
com.smartalarm.bms.command_changed.v1
com.smartalarm.bms.connection_changed.v1
com.smartalarm.bms.health_changed.v1
```

```
com.smartalarm.bms.map_published.v1
com.smartalarm.bms.point_observed.v1
com.smartalarm.bms.reconciliation_completed.v1
```

The typed event envelopes preserve provenance, mapping revision, source quality and freshness; missing source time remains unknown. Physical feedback and command execution retain the existing physical domain boundary. Large snapshots use object references and hashes.

Size against the source rather than a universal point count

```
poll_samples_per_second = Σ(N_group / poll_interval_seconds)
poll_requests_per_second = Σ(N_group / effective_batch_size / interval)
subscription_bytes_per_second = Σ(change_rate × payload_size) + protocol overhead
edge_journal_bytes >= peak_accepted_bytes_per_second × offline_seconds × overhead_factor
```

Planning example, not a benchmark:2,000 points every 5 seconds plus8,000 every 60 seconds produces about 533.3samples/s. At 300 bytes/sample this is160KB/s, approximately13.824GB for 24 hours before headers, indexes and replication; a factor of 2 yields 27.65GB, still requiring reserve and alarm-burst assessment. Snapshots, reconnects and shadow readers add separate load. Cloud autoscaling does not increase legacy-BMS capacity. Measure source→edge, edge commit→cloud, cloud→operator, request→native receipt and receipt→physical feedback separately. Publish p95/p99 only with accepted clock quality and workload fixtures.

5. Accept capabilities and hand over with evidence

Artifact releases move through draft→schema_validated→contract_tested→security_reviewed→approved→published→retired. Site activation moves through proposed→discovered→mapped→lab_verified→site_commissioning→site_accepted→active; healthy/degraded/offline/quarantined is a separate health dimension. Included JSON remains design_example. JSON Schema checks structure and selected guards, not signatures, evidence authenticity or controller authority.

An upgrade captures image/profile/mapping/policy digests, cursors and outstanding intents; validates compatibility; runs an authorized-budget shadow reader without ACK/write; drains new commands; persists checkpoints; transfers writer authority through fencing; and reconciles before reopening writes. Unknown physical outcomes or orphan leases keep writes disabled with local handover. Code/config rollback cannot undo acknowledgements or effects already performed. Retirement reviews open alarms/leases, transfers responsibility, revokes credentials/certificates and retains audit.

20 business cases are test specifications, not executed FAT/SAT

ID	Stimulus	Required outcome
BMS-T01	Rename, type, unit or enum changes	Keep identity for rename; quarantine incompatible semantics and affected commands.
BMS-T02	Repeated native ID across sites/tenants	No cross-tenant joins, disclosure or wrong acknowledgement.

ID	Stimulus	Required outcome
BMS-T03	Alarm returns normal before acknowledgement	Active becomes false; acknowledgement can remain pending; incident stays independent.
BMS-T04	Alarm recurs while an old ACK is pending	Old receipt cannot acknowledge the new occurrence; refresh and show conflict.
BMS-T05	Panel direct feed and BMS mirror replay100 times	One physical lineage; no duplicated incident, dispatch or broadcast.
BMS-T06	Lost cursor; paged snapshot with concurrent events	Record gaps; do not clear alarms based on absence from a single page.
BMS-T07	Higher-priority local override	Show rejected request or accepted non-winning demand accurately; no override bypass.
BMS-T08	Receipt lost after possible command execution	Mark unknown; reconcile without blind resending.
BMS-T09	Lease expires and slot owner changes during restart	Do not clear another owner; approved relinquishment requires readback.
BMS-T10	Fan demand accepted but proof false or bad	No physical success; show demand/feedback discrepancy.
BMS-T11	Link loss, full buffer and two-node failover	Local protection independent; durable commit and single-writer fencing hold.
BMS-T12	Alarm burst overlaps historian backfill	Alarm queue remains available and upstream budgets are respected.
BMS-T13	Expired/revoked certificate or wrong tenant	Privileged operations fail closed; no insecure fallback.
BMS-T14	Published incident is reflected by BMS	No recursive incident creation; preserve exact receipt evidence.
BMS-T15	Maintenance ends before restoration	Restoration remains pending and escalates; do not mark ready.
BMS-T16	Stale door/fan evidence affects route	Re-evaluate or withdraw route and old speech; assistance remains accessible.
BMS-T17	AI-derived point, occupancy count or old badge event	Do not present inference as measurement or confirmed occupant safety.
BMS-T18	VI/EN/JA, keyboard, zoom and color deficiency	Critical action/source owner readable; raw identifiers remain unchanged.
BMS-T19	Exercise data reconnects after outage	No exercise leak to actual operations; backfill cannot actuate.
BMS-T20	Retirement with open commands and leases	Handover, revoke, verify and retain audit; no orphan remote override.

28 protocol cases supplement and overlap the business cases

ID	Stimulus	Required outcome
BMS-P-01	Read-only enforcement — Attempt writes and external ACK with read-only identity	Reject before dispatch; no native write audit entry.
BMS-P-02	Scaling — Known pressure raw600 with scale100 and invalid unit	Decode6.0bar; unapproved mapping never enters rules.
BMS-P-03	Register addressing — Zero/one-based addresses and alternate32-bit word order	Match golden fixtures; mismatches block commissioning.
BMS-P-04	COV expiry — Subscription expiry or BMS restart	Renew and reconcile; report stale during gap.
BMS-P-05	MS/TP contention — Authorized burst on a constrained bus	No polling storm; record actual latency and critical-event handling.
BMS-P-06	Existing active alarms — Connect after alarms are already active	Load current conditions; stay syncing until reconciled.
BMS-P-07	Refresh race — New event arrives between refresh boundaries	Older refresh data cannot overwrite new state.
BMS-P-08	Duplicate delivery — Repeat identical source event	One canonical transition; multiple receipts retained.
BMS-P-09	Dual paths — Direct panel and BMS relay same alarm	Link physical lineage; preserve both paths.
BMS-P-10	Similar unrelated alarms — Equal labels and near timestamps on different devices	No false merge based only on text/time.
BMS-P-11	Late acknowledgement — Occurrence2 starts before ACK1 receipt	ACK1 never clears ACK2 responsibility.
BMS-P-12	Internal acknowledgement — SmartAlarm operator accepts work	Native BMS acknowledgement remains unchanged.
BMS-P-13	Return to normal — Source normal while rescue is unresolved	Incident and occupant safety remain independent.
BMS-P-14	Dispatch timeout — Controller might have acted but receipt is lost	Unknown outcome and reconciliation; no blind retry.
BMS-P-15	Expired offline command — Reconnect after command expiry	Do not replay; require a new authorized intent.
BMS-P-16	Local override — Operator obtains higher priority	Do not report physical success; show ownership conflict.
BMS-P-17	Lost writer — Kill edge between dispatch and receipt persistence	Fence standby and reconcile before writes resume.
BMS-P-18	Retained priority slot — Power loss while edge holds a BACnet priority	Measure controller behavior; require accepted local fallback.
BMS-P-19	Same-slot collision — Another application writes the same priority	Do not assume ownership or clear with NULL automatically.

ID	Stimulus	Required outcome
BMS-P-20	Certificate rotation — Expired, changed or revoked trust material	Fail closed for privileged actions; no automatic trust.
BMS-P-21	Quality loss — Bad/uncertain point or missing source time	No false normal/safe display; propagate uncertainty.
BMS-P-22	Clock drift — Source clock skew and late delivery	Keep both timestamps; do not overwrite newer occurrence.
BMS-P-23	Alarm flood — Alarm burst plus ordinary telemetry	Preserve transitions; only approved telemetry may coalesce.
BMS-P-24	Disk capacity — Journal near/full capacity	Actionable degradation and explicit gaps; no silent discard.
BMS-P-25	Reflection loop — Summary sent out and reflected back	Preserve origin; no new incident or command loop.
BMS-P-26	Translation — VI/EN/JA status and action terminology	Same operational meaning; preserve native codes.
BMS-P-27	Isolation — Cross-tenant credential, mapping or command	Reject and audit without topology or health-data disclosure.
BMS-P-28	Emergency independence — Cloud and edge both unavailable	Local protection remains independent under its own accepted design.

The 18 protocol-assurance controls map to the 40 FR requirements; they are not 58 independent features. Controls use BMS-CTRL IDs to avoid confusion with screens BMS01–08.

ID	Control	FR-BMS	UI
BMS-CTRL-01	Survey	01	BMS01
BMS-CTRL-02	Endpoint access	02	I02/I07
BMS-CTRL-03	Discovery/import	03–04	BMS01/F04
BMS-CTRL-04	Semantic mapping	05–07	BMS02/F04
BMS-CTRL-05	Point explorer	08–10	F07
BMS-CTRL-06	Alarm reconciliation	11/30	BMS03
BMS-CTRL-07	Multi-source correlation	13	O05/BMS03
BMS-CTRL-08	Internal ACK	14–15	O03
BMS-CTRL-09	External ACK	14	BMS03
BMS-CTRL-10	Incident writeback	16	BMS08
BMS-CTRL-11	Command approval	22–24	F09/F10
BMS-CTRL-12	Outcome evidence	25–26	F11/F12
BMS-CTRL-13	Override/relinquish	24/27	BMS04
BMS-CTRL-14	History/evidence	10/37	O21/F07

ID	Control	FR-BMS	UI
BMS-CTRL-15	Health	32–34	I05/BMS01
BMS-CTRL-16	Maintenance	18	F14/F18
BMS-CTRL-17	FAT/SAT	35–36	F15/F17
BMS-CTRL-18	Lifecycle	31/40	BMS05/BMS07

FAT uses fixtures/emulators and a profile-matched BMS sandbox; SAT uses an authorized site configuration. Each case records expected/actual/evidence/version/witness; unexecuted cases remain not_run. Proving read APIs does not require activating real firefighting equipment. Physical-control tests need approved impairment, cause-and-effect, local personnel and stop procedures. Package schema checks prove only valid JSON structure and disabled-by-default design configurations.

Integration into the existing baseline 3.0 document set

Document	Integrated content
BRD	Integrated: BR-BMS objectives for retained BMS investment, evidence-based rescue and governance; includes BMS owner, facility, OT security and integrator roles; measure coverage/gaps/ACK outcomes.
FRD	40 detailed FR-BMS and WF-34–37 integrated; WF-01–33 preserved; capability-specific acceptance traced.
Basic Design	Integrated: BMS domain, edge/conduits, authority matrix, queue priorities and recovery; reuse Command.
Detail Design	Integrated: entities/APIs/events, occurrence identity, cursor/epoch, priority ownership, unknown reconciliation and sizing.
UI/UX	Integrated eight new screen specifications and mockups and mapped I/F/S/O/Q/G changes; use topology, priority ladders, freshness and timelines.
Device / standards catalogs	Record exact build/interface/license/PICS; distinguish documented, tested and unknown capabilities.
API/Event schemas	Version additive extensions without silent breaks; reuse physical observation and Command types.
Proposal VI/EN	Integrated: BMS all-in-one, authority boundaries, fire/no-SOS scenario and qualification; 40 FR and eight integrated screens in the appendix.
Acceptance / deployment	Map 48 cases by scope without additive coverage claims; bind build/point map/owner/SLA/rollback/handover.

Customer-facing narrative integrated into the proposal

The BMS continues to manage the building; SmartAlarm adds coordinated rescue using building evidence. Customers retain building services, supervisors and existing operating procedures while connecting alarms, equipment location, route readiness and protective-system information to apps, speakers and responders. Measure

source coverage, time to awareness, people not yet reached and handover outcomes, rather than connected-point counts alone. Two-way functions are selected per interface and authority dossier, not sold as universal plug-and-play.

Fire on floor 12: The BMS forwards an alarm with source and time. SmartAlarm correlates a direct-panel feed without double counting, identifies the affected area and assesses routes against valid door/fan evidence. Assistance profiles and presence bring people who sent no SOS into the welfare-check queue. Apps/PA use approved messages; assigned responders acknowledge responsibility. Any approved BMS action passes through Command policy and edge enforcement, with receipt, controller report and feedback shown separately. Return to BMS normal does not close the incident until the incident commander verifies rescue outcomes.

Before estimating deployment, confirm product/build/license, interface contracts, point/alarm dictionaries, network boundaries, intended ACK/write authority, local cause-and-effect, scale/timing/offline requirements and SAT owners. The platform architecture is complete from the outset; evidence-gated activation is a quality gate, not a temporary demo architecture.

Source and artifact appendix

Research branches located/cross-checked sources on24 October 2026. Live specifications can change; archive approved versions during commissioning. The newer Modbus PDF URL timed out when opened; Modbus conclusions were cross-checked against Introduction/Specifications pages and vendor register documentation, not claimed as a complete PDF read. A source ledger is not site certification.

ID	Source	URL
S01	Haystack Ops	Open
S02	Brick Relationships	Open
S03	Brick External Representation	Open
S04	Haystack Points	Open
S05	BACnet Committee – PICS	Open
S06	BACnet International	Open
S07	JCI BACnet/SC overview	Open
S08	OPC Foundation – BACnet companion model	Open
S09	Beckhoff command prioritization	Open
S10	MSA BACnet MS/TP Driver Manual	Open
S11	CODESYS BACnetAcknowledgeAlarm	Open
S12	OPC UA Part 4 – DataValue	Open
S13	OPC UA Part 9 – acknowledgeable conditions	Open
S14	OPC UA Part 9 – Acknowledge	Open
S15	OPC UA Part 9 – synchronization	Open
S16	OPC UA Part 4 – Services	Open

ID	Source	URL
S17	Modbus Organization – introduction	Open
S18	Modbus Application Protocol V1.1b3	Open
S19	Schneider DOCA0302EN-04	Open
S20	Modbus Organization – specifications	Open
S21	OASIS MQTT 5.0	Open
S22	NIST publication status	Open
S23	NIST SP 800-82r3, phần network segmentation	Open
S24	REST specification	Open
S25	OPC specification	Open
S26	BACnet specification	Open
S27	Authentication and services	Open
S28	2021 application note	Open
S29	Official SmartConnector developer guide	Open
S30	API v6 reference	Open
S31	Metasys UI licensing	Open
S32	API v4 introduction	Open
S33	EBI current product	Open
S34	R600 guide specification, sections OPC/API	Open
S35	WEBs product	Open
S36	BTL Honeywell listings	Open
S37	Supervisor4.6 datasheet	Open
S38	2025 driver catalog	Open
S39	NDS product and prerequisites	Open
S40	Tridium marketplace: BTIB ActiveAPI	Open
S41	enteliWEB4.23 catalog, pages4/7	Open
S42	Official integration sheet2025	Open
S43	EC-Net security	Open
S44	EC-Net Facilities	Open
S45	Distech developer portal	Open

40-function CSV inventory · Source ledger CSV · Connector schema

Profile	JSON
bms.siemens.desigo-wsi	JSON
bms.schneider.ebo-ews	JSON
bms.jci.metasys-rest	JSON
bms.honeywell.ebi-api	JSON
bms.honeywell.webs-niagara	JSON
bms.tridium.niagara-local	JSON
bms.tridium.nds-cloud	JSON
bms.delta.enteliweb	JSON
bms.alc.webctrl-events	JSON
bms.alc.webctrl-soap-sdk	JSON
bms.distech.ecnet-niagara	JSON